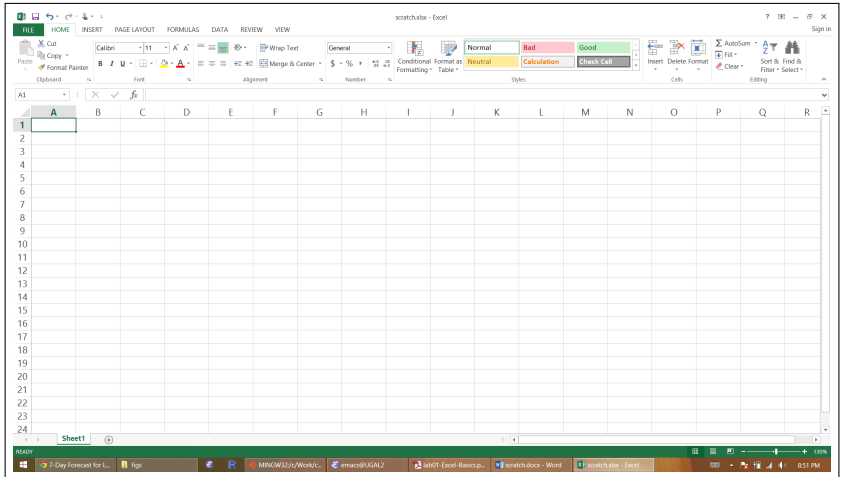
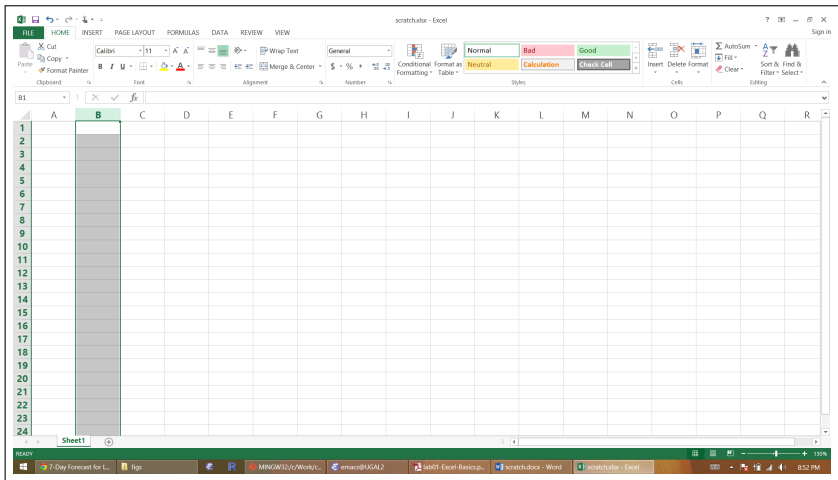


Applied Population Dynamics

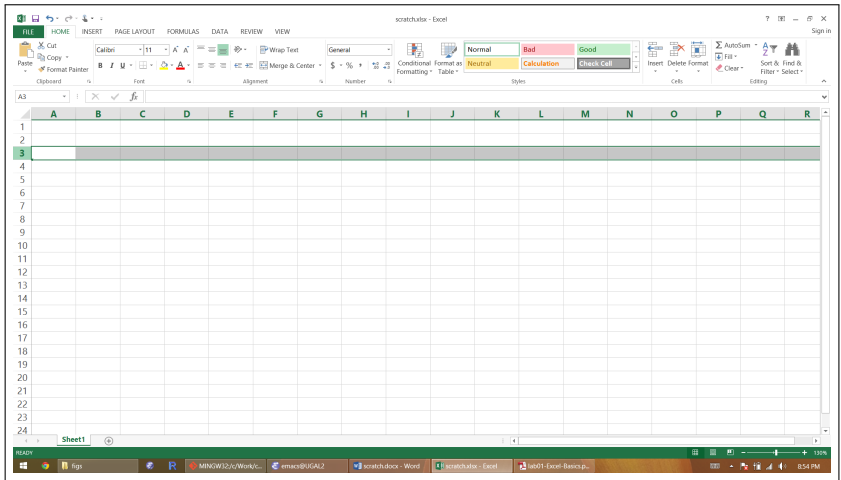
Lab 1 – Excel and R Basics



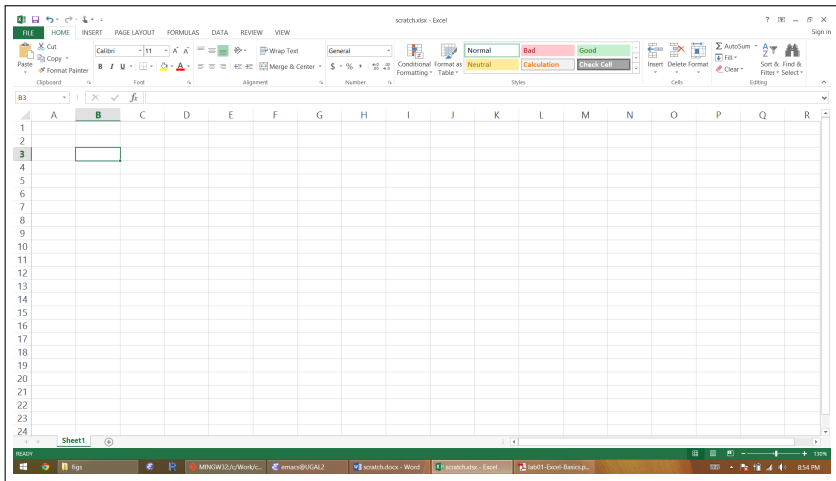
COLUMN B



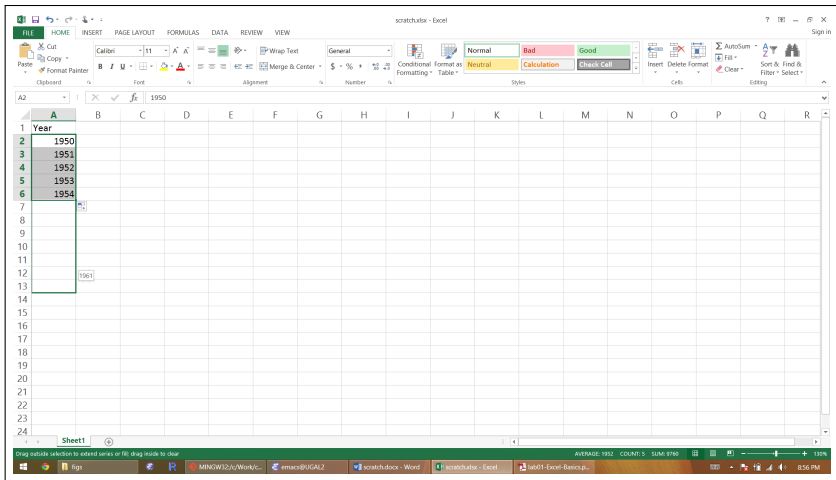
Row 3



CELL B3

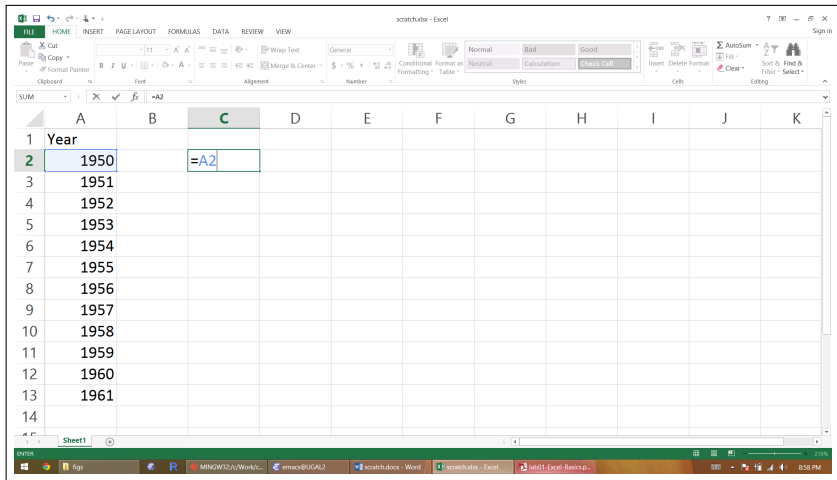


CREATE SEQUENCE USING AUTO-FILL



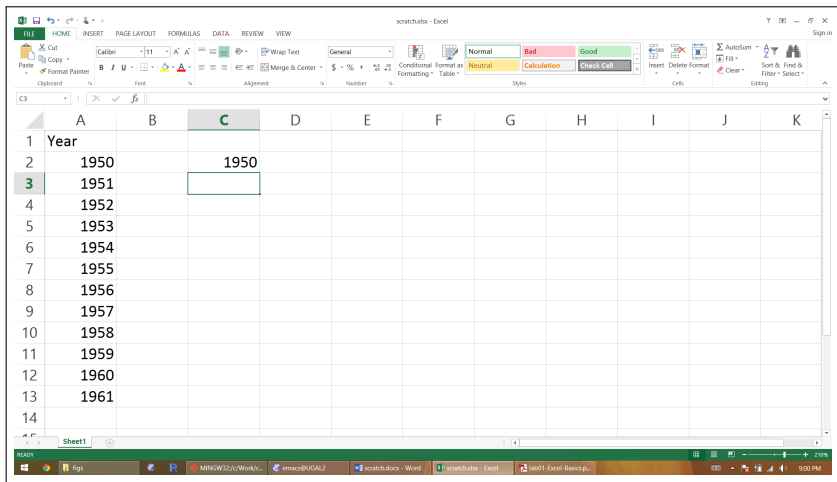
To use auto-fill: begin a sequence, highlight the cells, and then drag the box at the bottom-right of the last cell.

RELATIVE CELL REFERENCES



Cell referencing must begin with the “equals sign” (=)

RELATIVE CELL REFERENCES



Cell C2 will take on the value of A2

RELATIVE CELL REFERENCES

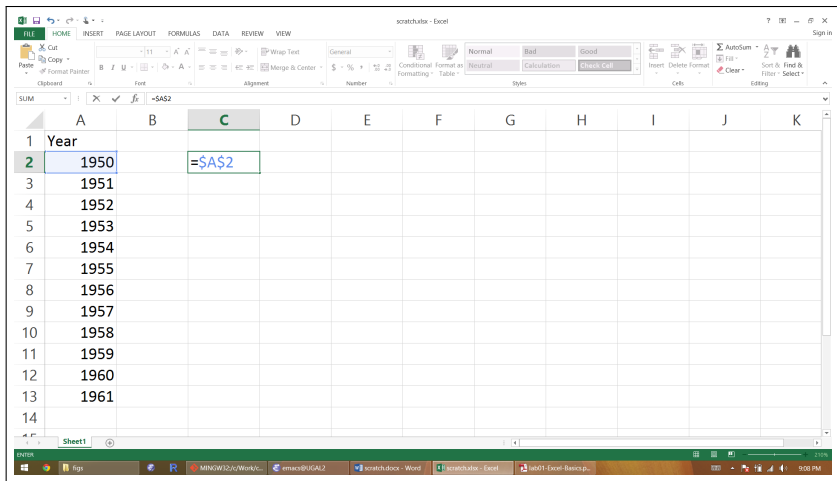
The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E	F	G	H	I	J	K
1	Year										
2	1950		1950								
3	1951		1951								
4	1952		1952								
5	1953		1953								
6	1954		1954								
7	1955		1955								
8	1956		1956								
9	1957		1957								
10	1958		1958								
11	1959		1959								
12	1960		1960								
13	1961		1961								
14											

The formula bar at the top shows the formula for cell C13 is `=A13`, demonstrating a relative cell reference.

Values of reference will change when using auto-fill

ABSOLUTE CELL REFERENCES



Dollar sign “locks” a reference so that auto-fill won’t change it

ABSOLUTE CELL REFERENCES

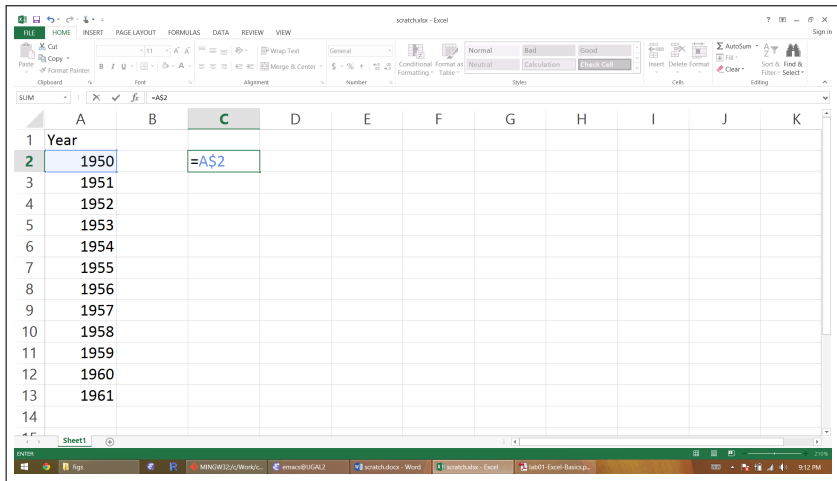
The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E	F	G	H	I	J	K
1	Year										
2	1950		1950								
3	1951		1950								
4	1952		1950								
5	1953		1950								
6	1954		1950								
7	1955		1950								
8	1956		1950								
9	1957		1950								
10	1958		1950								
11	1959		1950								
12	1960		1950								
13	1961		1950								
14											

The formula bar shows the formula in cell C2 is `=SAS2`. The status bar at the bottom indicates: AVERAGE: 1950, COUNT: 12, SUM: 23400, 210%.

Dollar sign “locks” a reference so that auto-fill won’t change it

PARTIAL CELL REFERENCES



The screenshot shows the Microsoft Excel interface with the following data in the spreadsheet:

	A	B	C	D	E	F	G	H	I	J	K
1	Year	Females									
2	1950	100									
3	1951	=B2 + B2*0.01									
4	1952										
5	1953										
6	1954										
7	1955										
8	1956										
9	1957										
10	1958										
11	1959										
12	1960										
13	1961										
14											

The formula bar at the top shows the active cell (B3) contains the formula: `=B2 + B2*0.01`.

Equations also have to begin with the equals sign (=)

scratch.xlsx - Excel

FILE HOME INSERT PAGE LAYOUT FORMULAS DATA REVIEW VIEW

Clipboard Font Alignment Number Styles

Normal Bad Good Neutral Calculation Check Cell

Insert Delete Format Clear Sort & Filter Select

AutoSum Fill

Sign in

B3 =B2 + B2*0.01

	A	B	C	D	E	F	G	H	I	J	K
1	Year	Females									
2	1950	100									
3	1951	101									
4	1952	102.01									
5	1953	103.03									
6	1954	104.06									
7	1955	105.101									
8	1956	106.152									
9	1957	107.214									
10	1958	108.286									
11	1959	109.369									
12	1960	110.462									
13	1961	111.567									
14											

Sheet1

READY AVERAGE: 106.2043728 COUNT: 11 SUM: 1168.25001 210%

figs MINGW32\z\Work\c... emacs@UGA2 scratch.docx - Word scratch.xlsx - Excel lab01-Excel-Basics.p...

9:16 PM

EQUATIONS

scratch.xlsx - Excel

FILE HOME INSERT PAGE LAYOUT FORMULAS DATA REVIEW VIEW

Clipboard Font Alignment Number Styles Cells

B2 =B2*0.8

	A	B	C	D	E	F	G	H	I	J	K
1	Year	Females	Males								
2	1950	100	=B2*0.8								
3	1951	101									
4	1952	102.01									
5	1953	103.03									
6	1954	104.06									
7	1955	105.101									
8	1956	106.152									
9	1957	107.214									
10	1958	108.286									
11	1959	109.369									
12	1960	110.462									
13	1961	111.567									
14											

Sheet1

ENTER

Windows taskbar: fips, MINGW32/c/Workc..., emacs@UGA2, scratchdock - Word, scratchdock - Excel, lab01-Excel-Basics.p...

9:18 PM

scratch.xlsx - Excel

FILE HOME INSERT PAGE LAYOUT FORMULAS DATA REVIEW VIEW

Clipboard Font Alignment Number Conditional Formatting Styles Cells Editing

Normal Bad Good Neutral Calculation Check Cell

AutoSum Fill Clear Sort & Find & Filter Select

C2: $=B2*0.8$

	A	B	C	D	E	F	G	H	I	J	K
1	Year	Females	Males								
2	1950	100	80								
3	1951	101	80.8								
4	1952	102.01	81.608								
5	1953	103.03	82.4241								
6	1954	104.06	83.2483								
7	1955	105.101	84.0808								
8	1956	106.152	84.9216								
9	1957	107.214	85.7708								
10	1958	108.286	86.6285								
11	1959	109.369	87.4948								
12	1960	110.462	88.3698								
13	1961	111.567	89.2535								
14											

Sheet1

READY AVERAGE: 84.35002009 COUNT: 12 SUM: 1014.600241

MINGW32\c\Workc... emacs@UGAU2 scratch.docx - Word scratch.xlsx - Excel ied01-Excel-Basics.p...

9:19 PM

FORMULAS

scratch.xlsx - Excel

FILE HOME INSERT PAGE LAYOUT FORMULAS DATA REVIEW VIEW

Clipboard Font Alignment Number Styles

Normal Bad Good Neutral Calculation Check Cell

AutoSum Sort & Find & Filter Select Clear

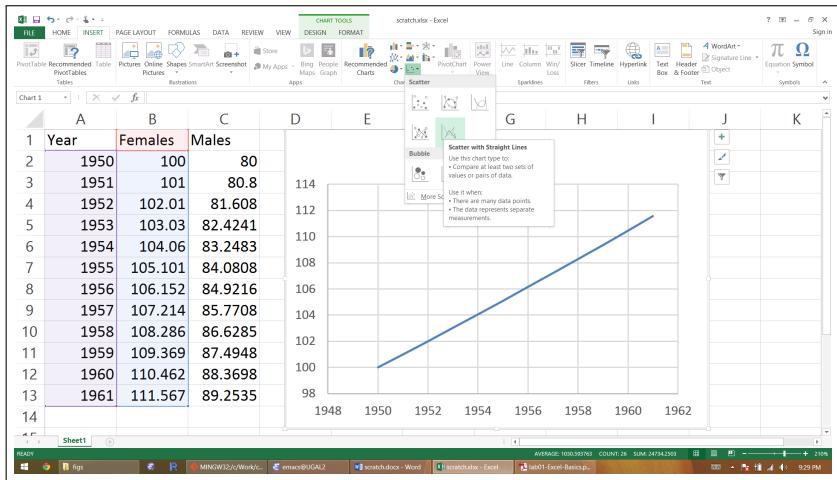
SUM X ✓ fx =ROUND(C4, 3)

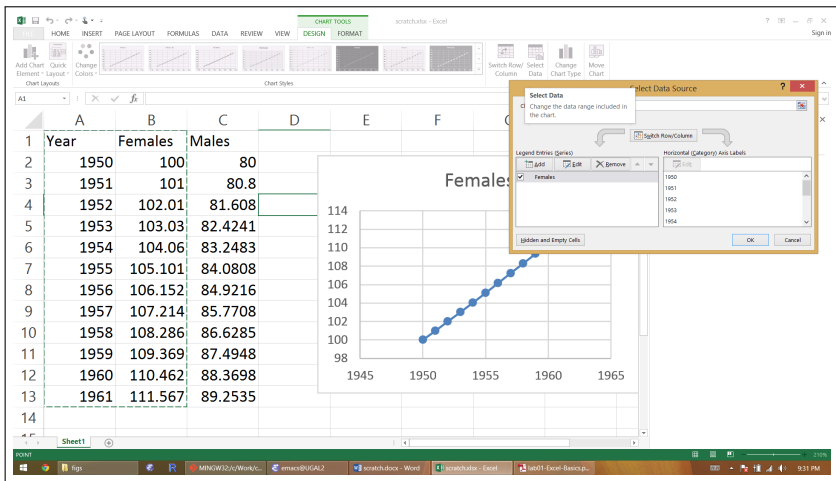
	A	B	C	D	E	F	G	H	I	J	K
1	Year	Females	Males								
2	1950	100	80								
3	1951	101	80.8								
4	1952	102.01	81.608	=ROUND(C4, 3)							
5	1953	103.03	82.4241								
6	1954	104.06	83.2483								
7	1955	105.101	84.0808								
8	1956	106.152	84.9216								
9	1957	107.214	85.7708								
10	1958	108.286	86.6285								
11	1959	109.369	87.4948								
12	1960	110.462	88.3698								
13	1961	111.567	89.2535								
14											

Sheet1

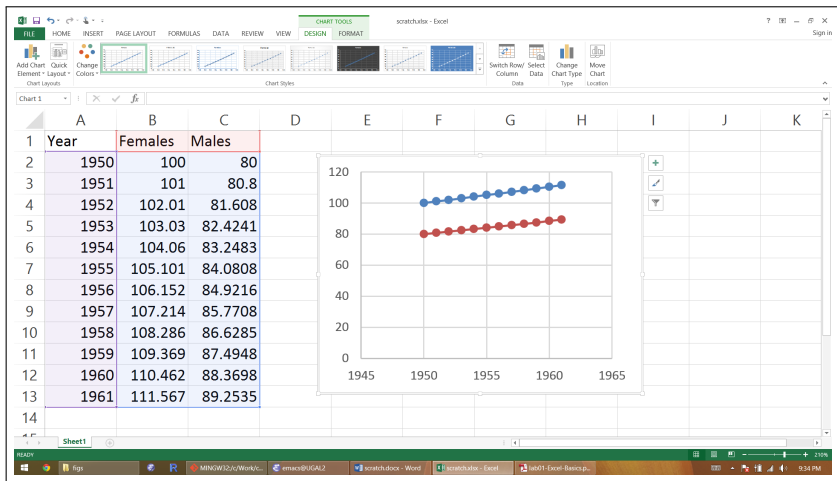
100%

9:21 PM

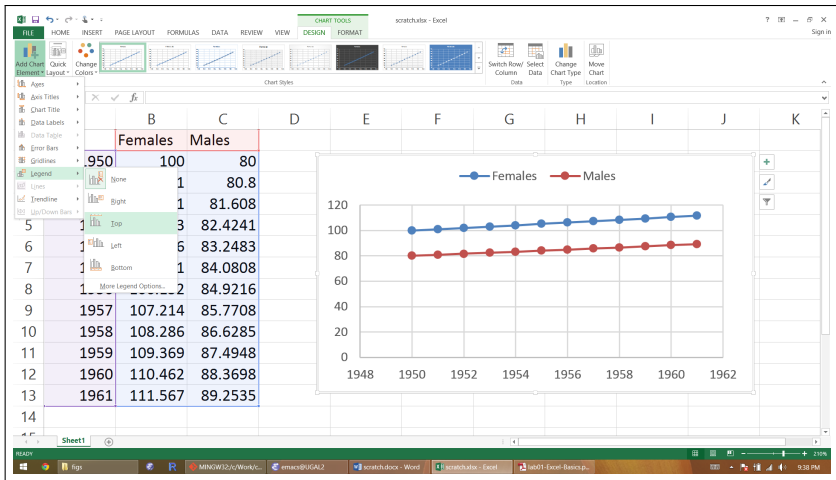




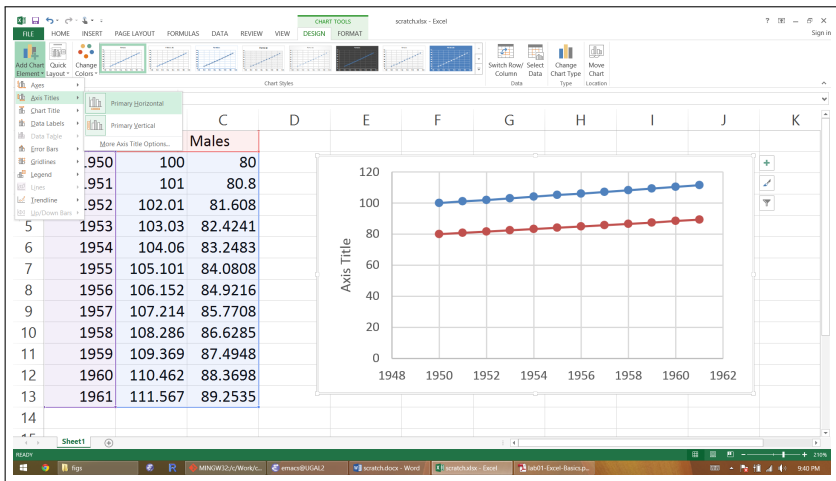
Add a line for males



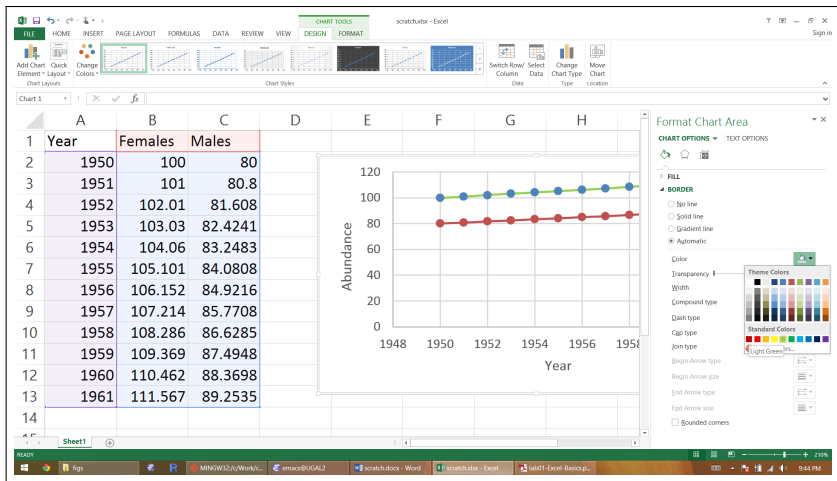
Add a line for males



Add legend



Add axis labels



Change line color

R can be downloaded here: <https://www.r-project.org/>

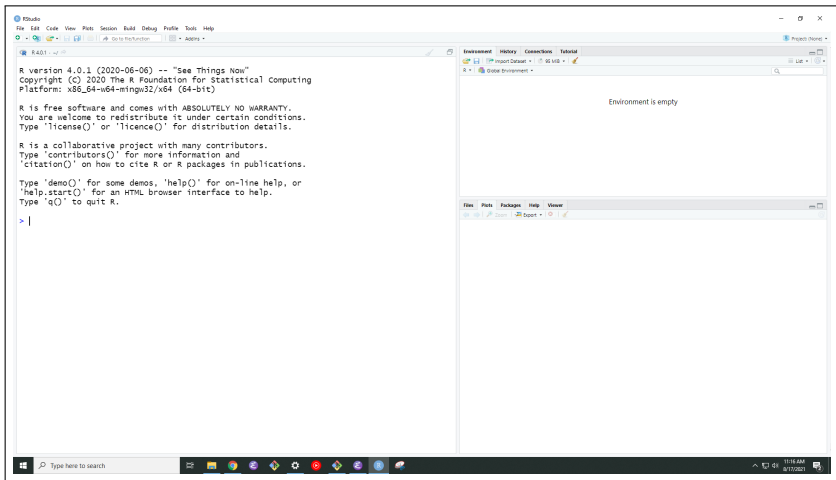
R can be downloaded here: <https://www.r-project.org/>

You can use the graphical user interface that comes with **R**, or you can run **R** through a system like **ESS+emacs** (<https://vgoulet.act.ulaval.ca/en/home/>) or **RStudio** (<https://www.rstudio.com/>)

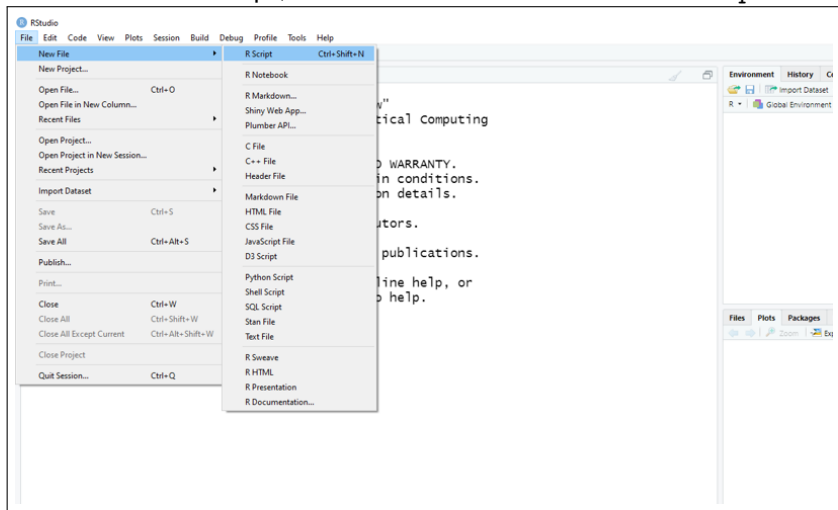
R can be downloaded here: <https://www.r-project.org/>

You can use the graphical user interface that comes with **R**, or you can run **R** through a system like **ESS+emacs** (<https://vgoulet.act.ulaval.ca/en/home/>) or **RStudio** (<https://www.rstudio.com/>)

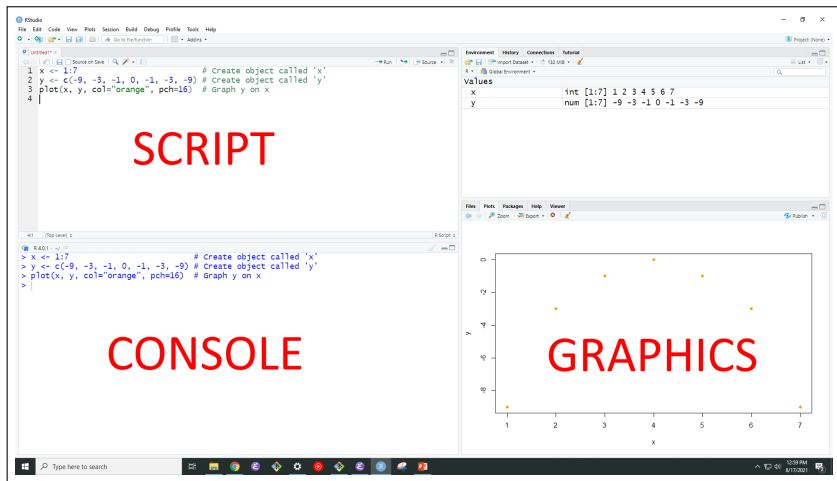
Most people use **RStudio** these days



To create a new script, click: File > New File > R Script



Save your script using: File > Save As



REPRODUCING THE EXCEL EXERCISE

Create an object called `year` to hold the sequence of years.

```
year <- 1950:1961 # A vector of integers
year            # Type the name of an object to see its values

## [1] 1950 1951 1952 1953 1954 1955 1956 1957 1958 1959 1960 1961
```

REPRODUCING THE EXCEL EXERCISE

Create an object called `year` to hold the sequence of years.

```
year <- 1950:1961 # A vector of integers
year           # Type the name of an object to see its values

## [1] 1950 1951 1952 1953 1954 1955 1956 1957 1958 1959 1960 1961
```

Note that you can use `seq` to create sequences in a more general way. The `c` function combines two or more vectors.

```
updown <- c(seq(from=20, to=100, by=20), seq(from=100, to=20, by=-20))
updown           # Two sequences, combined

## [1] 20 40 60 80 100 100 80 60 40 20
```

REPRODUCING THE EXCEL EXERCISE

Create an object called `year` to hold the sequence of years.

```
year <- 1950:1961 # A vector of integers
year           # Type the name of an object to see its values

## [1] 1950 1951 1952 1953 1954 1955 1956 1957 1958 1959 1960 1961
```

Note that you can use `seq` to create sequences in a more general way. The `c` function combines two or more vectors.

```
updown <- c(seq(from=20, to=100, by=20), seq(from=100, to=20, by=-20))
updown           # Two sequences, combined

## [1] 20 40 60 80 100 100 80 60 40 20
```

Use the `length` function to determine the number of values in a vector.

```
nYears <- length(year)
nYears

## [1] 12
```

A SIMPLE POPULATION MODEL

Create an empty vector to store the data on females. Set female abundance to 100 in the first year.

```
females <- rep(NA, nYears)
females[1] <- 100
```

A SIMPLE POPULATION MODEL

Create an empty vector to store the data on females. Set female abundance to 100 in the first year.

```
females <- rep(NA, nYears)
females[1] <- 100
```

Use a “for loop” to compute female abundance in subsequent years.

```
for(t in 2:nYears) {
  females[t] <- females[t-1] + females[t-1]*0.01
}
```

A SIMPLE POPULATION MODEL

Create an empty vector to store the data on females. Set female abundance to 100 in the first year.

```
females <- rep(NA, nYears)
females[1] <- 100
```

Use a “for loop” to compute female abundance in subsequent years.

```
for(t in 2:nYears) {
  females[t] <- females[t-1] + females[t-1]*0.01
}
```

We will use “for loops” for almost every population model that we implement in R

Generate the data on males
using a single line of code.

```
males <- females*0.8
```

A SIMPLE POPULATION MODEL

Put the objects in a `data.frame`

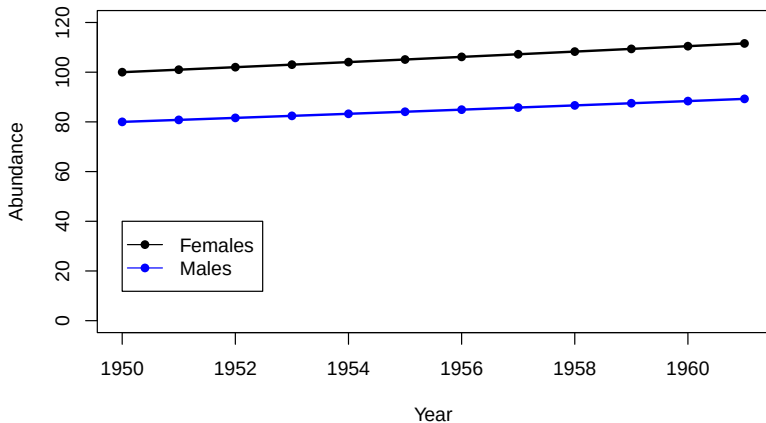
Generate the data on males
using a single line of code.

```
males <- females*0.8
```

```
modell1 <- data.frame(year, females, males)
modell1
```

##	year	females	males
## 1	1950	100.0000	80.00000
## 2	1951	101.0000	80.80000
## 3	1952	102.0100	81.60800
## 4	1953	103.0301	82.42408
## 5	1954	104.0604	83.24832
## 6	1955	105.1010	84.08080
## 7	1956	106.1520	84.92161
## 8	1957	107.2135	85.77083
## 9	1958	108.2857	86.62854
## 10	1959	109.3685	87.49482
## 11	1960	110.4622	88.36977
## 12	1961	111.5668	89.25347

```
plot(females ~ year, data=model1, type="o", xlab="Year", ylab="Abundance",
     lwd=2, pch=16, ylim=c(0, 120))
lines(males ~ year, data=model1, type="o", col="blue", lwd=2, pch=16)
legend(x=1950, y=40, legend=c("Females", "Males"), col=c("black", "blue"), lty=1, pch=16)
```



1. Create an Excel file and name it “Yourlastname_Yourfirstname”.
2. Create the sheet shown on the next page using the techniques covered in this lab.
 - ▶ Use auto-fill to create the first two columns.
 - ▶ For the “Adults” column, use the equation shown for cells C3 through C22. Note: For cell C2, you can directly enter the value “10”.
3. Copy “Sheet1” to a new sheet and change the color and thickness of the lines. You can pick any colors and thicknesses you want.
4. Replicate steps 1–3 using a “for loop” in a self-contained R script. Note that you will need code to create both graphs—the original one and the one with the new line color and thickness.

Hint: You can make the juveniles data using `seq` and `c`, like we did with the `updown` example earlier.

5. Upload the Excel workbook (with both sheets) and the R script to ELC by 5:00pm on Friday.

ASSIGNMENT

